
Price incorporation after scheduled macro releases

A measurement study of BTCUSDT

Estimand, model and validation

Research design. No primary fit has run.

Outline

Part 1: Measurement and model

1. Releases, prices and event time
2. Fraction of the move, response curve and half-time
3. Joint likelihood and partial pooling
4. Posterior inference and uncertainty

Part 2: Validation and interpretation

5. Pre-data checks and fit diagnostics
6. Robustness checks, outcome rules and limits

Motivation: separate speed from magnitude

Question

- How much of a release-window price move occurs in the first seconds?
- A large early move could mean a large response, a fast response, or both.
- CPI and employment news can move prices in either direction.

Design

- Anchor returns to a scheduled release time.
- Estimate the signed magnitude and the rate separately for each event.
- Pool event rates to estimate a population half-time.
- Test whether its 95% interval lies below 60 seconds.

The question concerns repricing, not return prediction or a trading rule.

Data: scheduled releases in a continuously traded market

Primary sample

- 106 CPI and 106 Employment Situation releases, scheduled at 08:30 ET.
- 1 September 2017 to 31 July 2026; boundaries set by archive availability.
- Dates from verified release calendars, not inferred from price jumps.

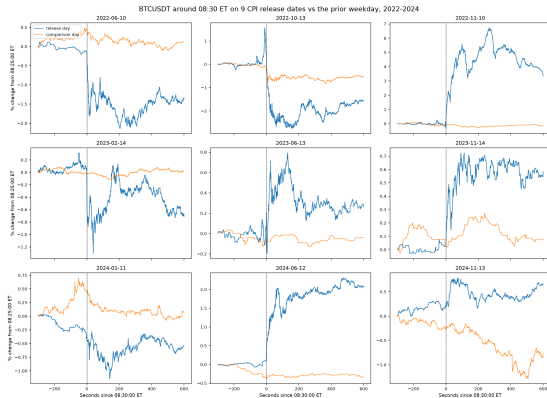
Prices and comparison sample

- Binance BTCUSDT spot, one-second bars, publicly downloadable with checksums.
- No market opening boundary at release time.
- 71 FOMC statements fitted separately: the press conference contaminates the hour.

212 is the calendar count before exclusions. The 2026-08-30 deviation corrects the preregistered 214.

Sources: ADRs 0001 to 0003; `release_calendar.csv`. Other-agency release screening remains open (#41).

Exploratory check: prices react at the release time

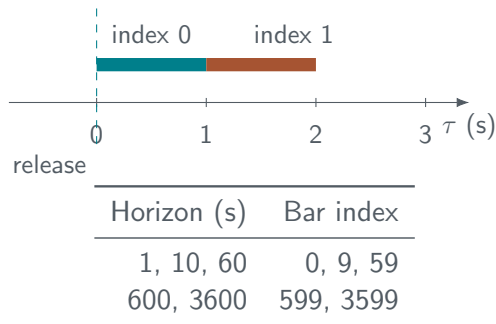


- Nine CPI dates, each compared with the prior weekday at the same clock time.
- This is not a half-time estimate or a representative sample.
- These dates also informed the rate prior; that reuse is disclosed.

Source: #6 and notebooks/cpi-kill-check.ipynb. The plot ends at 10 minutes, not one hour.

Event time: one second means one elapsed second

- Release e occurs at t_e ; event time is $\tau = t - t_e$, in seconds.
- $P_e(0^-)$ is the last traded price strictly before t_e .
- ET dates are converted to UTC using the offset on that date.
- A bar opening at index 0 covers the first post-release second.



ADR 0004: returns use the close of bar $\tau - 1$; the likelihood uses elapsed time τ , never the bar index.

Observed returns and the modelled response

$$R_e(\tau) = \log P_e(\tau) - \log P_e(0^-),$$

$$R_e(\tau) = \underbrace{m_e(\tau)}_{\text{modelled mean}} + \underbrace{\varepsilon_e(\tau)}_{\text{background movement}}.$$

- $m_e(\tau)$ is the expected return conditional on event e 's parameters, not an average of up and down releases.
- Observe cumulative returns at $\mathcal{T} = \{1, 10, 60, 600, 3600\}$ seconds.
- $H = 3600$ s is the terminal horizon; $M_e := m_e(H)$ is the signed terminal magnitude.
- Log returns are dimensionless; $10^4 R_e$ expresses them in basis points (bp).

The mean is a model component, not an identified causal effect of the release alone.

The estimand: fraction of the expected terminal move

$$\phi_e(\tau) := \frac{m_e(\tau)}{m_e(H)} = \frac{m_e(\tau)}{M_e}, \quad M_e \neq 0.$$

What it measures

- The expected fraction reached by τ , relative to the one-hour destination.
- Magnitude cancels; a falling response can have the same fraction as a rising one.
- $\phi_e(H) = 1$ by definition, not as an empirical finding.

ADR 0002; docs/framings/fraction-of-move.md.

What it does not measure

- The fraction of an observed return, $R_e(\tau)/R_e(H)$.
- Distance to an independently observed efficient price.
- Speed identified from a zero response: at $M_e = 0$, the data contain no rate information.

Why averaging observed ratios fails

For E events in the fitted sample,

$$\hat{\phi}(\tau) = \frac{1}{E} \sum_{e=1}^E \frac{R_e(\tau)}{R_e(H)} \quad \text{is not the chosen estimand.}$$

Noisy denominator

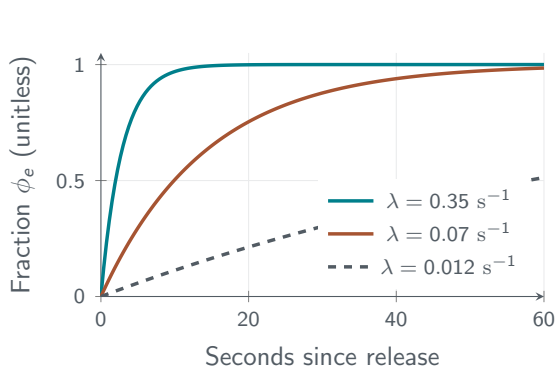
- A one-hour return near zero produces an arbitrarily large ratio.
- Averaging does not remove this instability; the ratio's expectation need not exist.

Selection is not a repair

- Dropping small one-hour returns selects on a post-release outcome.
- Fit magnitude and rate jointly instead; retain every event meeting the data rules.

Avoiding division by data removes this estimator's failure. It does not guarantee that every event's rate is identified.

Response curve: one magnitude and one rate per event



$$m_e(\tau) = M_e \frac{1 - e^{-\lambda_e \tau}}{1 - e^{-\lambda_e H}}$$

- $\lambda_e > 0$ is the rate, in s^{-1} : larger means faster.
- Normalisation ensures $m_e(H) = M_e$, not merely an asymptotic limit.
- One monotone response: the form cannot represent overshoot or reversal.

Curves illustrate the model at fixed rates; they are not fitted results.

Half-time: solve for half the terminal response

$$\begin{aligned}\phi_e(\tau_{1/2}) &= \frac{1}{2} \\ 1 - e^{-\lambda_e \tau_{1/2}} &= \frac{1}{2}(1 - e^{-\lambda_e H}) \\ e^{-\lambda_e \tau_{1/2}} &= \frac{1}{2}(1 + e^{-\lambda_e H}) \\ \tau_{1/2}(\lambda_e) &= \frac{1}{\lambda_e} \log \frac{2}{1 + e^{-\lambda_e H}}.\end{aligned}$$

- Half-time is in seconds and decreases as the rate increases.
- If $\lambda_e H$ is large, $\tau_{1/2} \approx \log 2 / \lambda_e$.
- As $\lambda_e \rightarrow 0$, the normalised path becomes linear and half-time tends to $H/2$.
- Below one second, a half-time is an extrapolation from the curve, not resolved timing.

Likelihood: the five returns are one observation vector

$$\mathbf{R}_e = (R_e(\tau_1), \dots, R_e(\tau_5))^\top \sim \mathcal{N}(\mathbf{m}_e, \Sigma_e), \quad \Sigma_{e,jk} = \varsigma_e^2 \min(\tau_j, \tau_k).$$

- $\mathbf{m}_e$ contains the response curve at the five horizons.
- ς_e is event-specific background volatility, in return units per $\sqrt{s}$.
- The covariance assumes zero-mean, independent background increments with variance proportional to elapsed time.
- Different events are conditionally independent given their parameters and the population parameters.

These are assumptions of the likelihood. Jumps, changing within-hour volatility and baseline noise can violate them.

Nested horizons share the same early increments

For the first three horizons:

$$\Sigma_e = \varsigma_e^2 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 10 & 10 \\ 1 & 10 & 60 \end{pmatrix}.$$

- Diagonal: variance at each elapsed time.
- Off-diagonal: variance accumulated during the shared interval.

Write the 60-second noise as

$$\varepsilon_e(60) = \varepsilon_e(10) + \Delta_e(10, 60),$$

where the later increment is independent.
Then

$$\begin{aligned} \text{Cov}\{\varepsilon_e(10), \varepsilon_e(60)\} \\ = \text{Var}\{\varepsilon_e(10)\} = 10\varsigma_e^2. \end{aligned}$$

- A diagonal covariance incorrectly treats the shared movement as fresh evidence.

Equivalent implementation: independent return increments, with variances $\varsigma_e^2(\tau_k - \tau_{k-1})$, $\tau_0 = 0$.

Partial pooling: event rates come from one population

$$\log \lambda_e \sim \mathcal{N}(\mu, \sigma^2), \quad \log \lambda_e = \mu + \sigma z_e, \quad z_e \sim \mathcal{N}(0, 1).$$

Population parameters

- μ : centre of log rates; $\exp(\mu)$ is the population median rate.
- σ : between-event spread of log rates, not posterior uncertainty about μ .
- The second expression is the non-centred form used for sampling.

Why pool?

- Separate rates allow events to differ.
- Weak events borrow information from the population rather than being discarded.
- Strong shrinkage is not evidence that a weak event has a well-measured rate.

Rate priors: $\mu \sim \mathcal{N}(-2.7, 1.5^2)$ and $\sigma \sim \text{HalfNormal}(1)$, with time in seconds.

How pooling weights an event's evidence

Local Gaussian approximation

Let $\theta_e = \log \lambda_e$, and suppose

$$\hat{\theta}_e \mid \theta_e \sim \mathcal{N}(\theta_e, s_e^2), \quad \theta_e \sim \mathcal{N}(\mu, \sigma^2).$$

With μ, σ fixed, the posterior mean is

$$\mathbb{E}[\theta_e \mid \hat{\theta}_e] = w_e \hat{\theta}_e + (1 - w_e) \mu,$$

$$w_e = \frac{\sigma^2}{\sigma^2 + s_e^2}.$$

The formula explains the weighting mechanism; it is not a replacement likelihood.

- Small event uncertainty s_e : more weight on the event estimate.
- Large s_e : more weight on the population centre.
- Larger population spread σ : less shrinkage.
- The actual nonlinear fit estimates event and population parameters jointly.

Priors: the specification is not yet ready for estimation

Preregistered

- Log-normal rate hierarchy with the priors on the preceding slides.
- $\log |M_e| \sim \mathcal{N}(\mu_M, \sigma_M^2)$, but “sign free” is not a complete signed prior.
- $\log \varsigma_e \sim \mathcal{N}(\nu, \omega^2)$; volatility varies across events.

Decisions still open

- Consistent model units and a continuous signed-magnitude prior (#31).
- Volatility priors and the prior-predictive bound, informed by candidate simulations (#42).
- Residual and robustness definitions (#43); concurrent-release treatment (#32).

Proposals are not adopted specifications. ADRs and a dated deviation (#44) must precede implementation and real-data fitting.

Posterior inference: sample the joint uncertainty

$$p(\Theta \mid \mathbf{R}) \propto \underbrace{p(\Theta)}_{\text{priors and hierarchies}} \prod_{e=1}^E \underbrace{p(\mathbf{R}_e \mid \Theta)}_{\text{joint likelihood for event } e}.$$

- Θ collects all event and population parameters. One posterior draw gives one value for each.
- The rate enters nonlinearly; numerical sampling replaces analytic posterior integration.
- NumPyro's No-U-Turn Sampler (NUTS) uses gradients to explore the posterior and chooses a trajectory length for each draw.
- Four chains start separately: 1000 warmup iterations, then 2000 retained draws each; acceptance target 0.8.

Warmup adapts the sampler and is discarded. The retained draws are correlated; diagnostics assess their numerical reliability.

Posterior summaries: transform every draw, then take quantiles

For retained draw s , compute

$$\lambda_{\text{pop}}^{(s)} = e^{\mu^{(s)}}, \quad t_{\text{pop}}^{(s)} = \frac{1}{\lambda_{\text{pop}}^{(s)}} \log \frac{2}{1 + e^{-\lambda_{\text{pop}}^{(s)} H}}.$$

- Report the median and the 2.5th and 97.5th percentiles of $t_{\text{pop}}^{(s)}$.
- This is the posterior of the population median half-time, not the mean of estimated event half-times.
- Similarly, report $\phi(\tau; e^{\mu^{(s)}})$ at each horizon, plus μ and σ , with 95% credible intervals.
- A credible interval describes parameter uncertainty under the model; it is not the spread of event half-times.

The population curve uses the median rate. It is not $\mathbb{E}_{\lambda}[\phi(\tau; \lambda)]$, the mean curve over events.

Part 2: Validation and interpretation

Pre-data gates, diagnostics, robustness and permitted claims

Before real-data fitting: plausibility and recovery

Prior predictive

- Draw population parameters, then event parameters, then returns.
- Preregistered criterion: at least 95% of $|R(H)|$ below 500 bp.
- The bound's justification needs correction: the exploratory grid was not a one-hour sample.
- Candidate priors and bounds are assessed before a choice is logged.

Recovery

- Simulate 100 datasets of 212 events and fit each with the declared settings.
- Count intervals containing the generating population half-time.
- Pass band: 90 to 99 of the 100 intervals contain the truth.
- Report failed fits and divergences too; coverage alone cannot diagnose their cause.

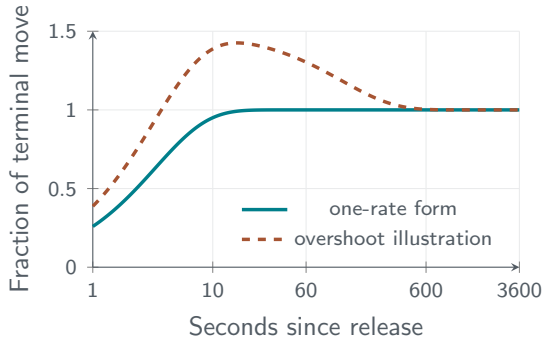
A failed gate blocks real-data fitting. Repair requires a recorded decision and deviation, then both checks are rerun.

Fit diagnostics: did sampling characterise the posterior?

Diagnostic	What it checks	Preregistered criterion
Rank-normalised split $\hat{R}$	Agreement across chains and their halves	$\hat{R} < 1.01$
Bulk and tail ESS	Effective information in correlated draws, in the centre and tails	Both > 400
Divergences	Numerical failures along sampling trajectories; possible biased exploration	Zero

- Criteria apply to every parameter as preregistered; proposed scope and tolerance changes remain decisions in #31.
- A rejected fit is reported as a study failure, not silently retuned. Passing is necessary, not proof of model validity.

Residual shape: can one rate describe the response?



- Raw residual: $R_e(\tau_k) - m_e(\tau_k)$ for each event and draw.
- Scaling, direction alignment and pooling determine which patterns remain visible.
- ADR 0006 must fix that diagnostic and a numerical overshoot rule before it is applied.
- Systematic overshoot invalidates a single-rate interpretation; the finding is about shape.

The dashed curve illustrates a limitation, not the pending two-component specification or an estimated residual pattern.

Robustness: horizon, functional form and baseline

Check	Change	Question
1. Terminal horizon	$H = 30$ min and $H = 4$ h	Does the chosen destination drive the half-time?
2. Functional form	Add an overshoot-capable response	Does it improve held-out predictive density?
3. Baseline	Mean price over the preceding 10 s	How sensitive is the curve to the last pre-release trade?

- The four-hour arm admits later scheduled news; it cannot establish isolated-release timing.
- Horizon grids, the alternative model and predictive comparison are fixed in ADR 0006, not chosen after fitting.

Robustness: population structure and prior sensitivity

Check	Change	Interpretation
4. Release type	Allow different log-rate centres	CPI versus Employment Situation; secondary
5. Time	Linear year trend in log rate	Change across sample years; secondary
6. Rate prior	$\mu \sim \mathcal{N}(0, 3^2)$	Sensitivity to reuse of exploratory dates
7. FOMC	Separate statement sample	Contaminated by the press conference

- Wide subgroup or trend intervals are reported as wide, not read as evidence of no difference.
- All seven checks remain planned. A concurrent-release refit requires a deviation (#32, #44).

Outcome rules: failures precede the half-time claim

1. **Fit rejected:** the posterior is not characterised. Report the diagnostic failure; no rate claim.
2. **Overshoot rule fires:** one rate does not describe the path. Report the shape failure; no rate claim.
3. **Otherwise:** classify the 95% interval $[L, U]$ for the population half-time.

Supported

$$U < 60 \text{ s}$$

Interval entirely below the threshold.

Inconclusive

$$L \leq 60 \leq U$$

Interval includes the threshold.

Falsified

$$L > 60 \text{ s}$$

Interval entirely above the threshold.

Publish null and inconclusive outcomes with the same prominence. Robustness arms qualify the primary verdict, not replace it.

Interpretation limits

- **Attribution:** other news can enter the window. A fitted response does not identify the effect of CPI or employment news alone.
- **Resolution:** one-second bars and an unverified exchange-clock offset limit the shortest measurable times.
- **Shape:** the half-time is conditional on the exponential form and the chosen terminal horizon.
- **Weak events:** a small magnitude leaves little rate information; a concentrated hierarchical posterior can be driven by pooling or priors.
- **Scope:** one asset, one venue and primary releases at 08:30 ET. No inference about all markets follows.

No empirical precision or gate outcome is claimed here. Full objections and planned checks: [docs/limitations.md](#).

What a completed measurement must report

- Population half-time and incorporation curve with 95% intervals, only where the outcome rules permit interpretation.
- Exclusion counts and reasons; pre-data gate results; fit diagnostics; residual verdict.
- Every planned robustness result, including rejected fits and analyses not run.
- Deviations from the original plan, with their reasons and the claims they change.

Current stopping point

The estimand is fixed. Specification decisions and validation gates remain open. No primary half-time has been estimated.

Audit trail: PREREGISTRATION.md, ADRs 0001 to 0004, and issues #31, #32, #42 to #45.

Backup: preregistered exclusions

An event is excluded only for a declared data failure:

1. The daily archive is missing or fails checksum verification.
 2. Fewer than 95% of seconds in $[-300, +3600]$ carry a trade.
 3. No trade occurs in the 60 seconds before release.
- No exclusion for a small terminal move, unexpected sign, implausible rate or poor fit.
 - Missing grid-second handling is still to be fixed in ADR 0006.
 - Concurrent-release treatment is a separate pending decision; any exclusion requires a logged sample change.

Backup: the original prior table and its unresolved parts

Parameter	Preregistered distribution	Status
μ	$\mathcal{N}(-2.7, 1.5^2)$	Rate prior retained
σ	HalfNormal(1)	Rate spread retained
μ_M	$\mathcal{N}(\log 30, 1^2)$	Intended magnitude: bp
σ_M	HalfNormal(1)	Incomplete sign prior
ν	$\mathcal{N}(0, 2^2)$	Volatility location
ω	HalfNormal(1)	Volatility spread

- $\log |M_e| \sim \mathcal{N}(\mu_M, \sigma_M^2)$ and $\log \varsigma_e \sim \mathcal{N}(\nu, \omega^2)$.
- The table's numerical scales and dimensionless returns are not consistent without an explicit unit conversion. ADR 0005 must resolve this and supply a signed prior.

Original plan reproduced for audit, not presented as an executable model. Candidate changes are not adopted by this deck.

Sources and further detail

- **Analysis contract:** PREREGISTRATION.md, including its dated sample-count deviation.
- **Model and notation:** docs/framings/; **limits:** docs/limitations.md.
- **Release and alignment decisions:** docs/adr/0001 to 0004.
- **NUTS:** Hoffman and Gelman (2014), *The No-U-Turn Sampler*, JMLR 15, 1593 to 1623.
- **Diagnostics:** Vehtari et al. (2021), *Rank-Normalization, Folding, and Localization*, Bayesian Analysis 16(2), 667 to 718.